

Ex 1 package com;  
import java.util.HashSet;  
public class SetOperation {  
 Psvm {

HashSet<Integer> nums = new HashSet<Integer>();

nums.add(10);

nums.add(20); [∴ doesn't allow duplicates] Output: [20, 10, 15]

nums.add(15);

nums.add(20); \* syso(nums); [20, 10, 15]

\* for (Integer num : nums) { output:  
 syso(num);      ⇒      20  
 10  
 15  
 }

\* Iterator<Integer> iterator = nums.iterator();

while (iterator.hasNext()) {

syso(iterator.next());

Output:

20

10

15

class: 34

}

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Ex 2 package com;  
public class Remove Duplicates {

public static void main(String[] args) {

ArrayList<Integer> nums = new ArrayList<Integer>();

Arrays.asList(10, 10, 20, 25, 30, 30);

ArrayList<Integer> result = new ArrayList<Integer>();

for (Integer num : nums) {

if (!result.contains(num)) {

result.add(num);

output:

[10, 20, 25, 30]

}

syso(result);

}

}



Ex: package com;

public class TreeSet Operation {

// TreeSet: Psum {

\* TreeSet<Integer> nums = new TreeSet<Integer>();

nums.add(10);

nums.add(20);

nums.add(4);

nums.add(12);

syso(nums);

output: [4 10 12 20]

\* set<String> names = new TreeSet<String>();

names.add("ABC");

names.add("AAA");

names.add("BBB");

syso(names);

output:

[AAA, ABC, BBB]

\* [We can pass inputs as parameters also, using Arrays.asList();]

// Linked HashSet

\*

Set<Integer> numslinked = new LinkedHashSet<Integer>();

numslinked.add(10);

numslinked.add(20);

numslinked.add(30);

numslinked.add(4);

numslinked.add(12);

syso(numslinked);

\* (Maintains Insertion Order)

output:

[10 20 30 4 12]

\*\*\*

\* In Integer & String internally sets override hashCode & equals() methods, due to this set doesn't allow duplicates.



```

package com;

public class HashSetOperation {
    Psum {
        set < Employees > emp = new Hash TreeSet < Employees > ();
        emp.add (new Employees (1, "AAA"));
        emp.add (new Employees (2, "BBB"));
        emp.add (new Employees (2, "BBB"));
        syso (emp);
    }
}

```

\* It uses HashCode & equals method to check the values are same & not

[Employee(2, 3) → hashcodes are different, due to this set doesn't recognize, due to this it prints duplicates also, to ignore this we have to override the hashCode & equals ().]

output:

Employees  
 id=1, name=AAA  
 id=2, name=BBB  
 id=2, name=BBB

\* Generate hashCode () & equals ().

output:

Employees  
 id=1; name=AAA  
 id=2; name=BBB

```

class Employees {
    int id;
    String name;

    public Employee () {
    }

    public Employees (int id, String name) {
        this.id = id;
        this.name = name;
    }
}

```

Getters & Setters.

To String ();

}

}